

SYSTEM DESIGN Important Technical Questions

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Backend Technical Interview Questions

1. What is difference between API Gateway and Load Balancer?
2. What is difference between Reverse Proxy and Forward Proxy?
3. What is difference between Horizontal scaling and vertical scaling?
4. What is difference between Microservices and Monolithic architecture?
5. What is difference between vertical and horizontal partitioning?
6. What is Rate Limiter? How does it work?
7. How does Single Sign On (SSO) works?
8. How does Apache Kafka works? Why is it so fast?
9. Difference between Kafka, ActiveMQ, and RabbitMQ?
10. Difference between JWT, OAuth, and SAML?
11. What is difference between SQL and NoSQL databases?
12. How does Content Delivery Network (CDN) work?
13. What is difference between Synchronous and Asynchronous communication?
14. How does Database Sharding work?
15. What is difference between REST and GraphQL?
16. How does Caching work? What are caching strategies?
17. What is difference between Strong and Eventual Consistency?
18. How does Message Queue work?
19. What is difference between TCP and UDP?
20. How does Database Replication work?
21. What is difference between Stateful and Stateless architecture?
22. How does Circuit Breaker pattern work?
23. What is difference between Authentication and Authorization?
24. How does Distributed Tracing work?
25. What is difference between ACID and BASE properties?
26. How does Service Discovery work?
27. What is difference between Push and Pull architecture?
28. How does Database Indexing work?
29. What is difference between Batch and Stream processing?
30. How does Consistent Hashing work?
31. What is difference between RPC and REST?
32. How does Leader Election work in distributed systems?
33. What is difference between Optimistic and Pessimistic locking?
34. How does Two Phase Commit work?
35. What is difference between CAP theorem components?
36. How does Bloom Filter work?
37. What is difference between WebSocket and HTTP?

- 38. How does Service Mesh work?
- 39. What is difference between Blue Green and Canary deployment?
- 40. How does Distributed Cache work?

A promotional graphic for the LLD Live Cohort 2.0. The background is a dark, blurred cityscape at night with bokeh light effects. In the upper right, a stylized rocket ship with orange and white segments is launching, leaving a trail of orange and yellow sparks. The main text is centered and reads "LLD Live Cohort 2.0 Launched!" in a large, bold, sans-serif font. "LLD" is white, "Live Cohort" is blue, and "2.0 Launched!" is yellow. Below the title, there is a list of three bullet points, each preceded by a yellow checkmark icon. The first bullet point is "Build Real-World System Design Projects". The second bullet point is "Weekend Live Classes" followed by "Tailored for Working Professionals" on the next line. The third bullet point is "Mock Interviews & Priority Career Mentorship". At the bottom of the graphic, a yellow banner contains the text "Limited Seats – Enroll Now! Register by 28 February" in a bold, sans-serif font.

LLD
Live Cohort
2.0 Launched!

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- ✓ Weekend Live Classes
Tailored for Working Professionals
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Limited Seats – Enroll Now! Register by 28 February

I've launched LLD Live Cohort 2.0 do check to enroll if you need structured guidance and mentoring:

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